

Atomic Structure

MCQ

1. Number of elements identified so far is

- A) 92
- B) 108
- C) 118
- D) 120

Answer: C

2. Elements occurring naturally are

- A) 26
- B) 92
- C) 118
- D) 50

Answer: B

3. Smallest particle of an element is

- A) Molecule
- B) Atom
- C) Cell
- D) Ion

Answer: B

4. The word atom is derived from

- A) Latin
- B) Greek
- C) French
- D) German

Answer: B

5. Atom means

- A) Divisible
- B) Large
- C) Indivisible
- D) Reactive

Answer: C

6. Structural and functional unit of matter is

- A) Molecule
- B) Atom
- C) Cell
- D) Compound

Answer: B

7. Atomic structure deals with

- A) Molecules only
- B) Elements only
- C) Structure of atom
- D) Chemical reactions

Answer: C

8. Centre of atom is called

- A) Orbit
- B) Shell
- C) Nucleus
- D) Cell

Answer: C

9. Positively charged particles are

- A) Electrons

B) Neutrons

C) Protons

D) Ions

Answer: C

10. Negatively charged particles are

A) Protons

B) Neutrons

C) Electrons

D) Nuclei

Answer: C

11. Neutral particles are

A) Protons

B) Electrons

C) Neutrons

D) Ions

Answer: C

12. Dalton proposed atomic theory in

A) 1774

B) 1779

C) 1808

D) 1904

Answer: C

13. Dalton's theory states atoms are

A) Divisible

B) Invisible

C) Indivisible

D) Charged

Answer: C

14. Atoms rearrange during

A) Physical change

B) Chemical reaction

C) Nuclear reaction

D) Evaporation

Answer: B

15. Dalton's theory explains law of

A) Motion

B) Gravitation

C) Conservation of mass

D) Electricity

Answer: C

16. Atoms of same element with different mass are

A) Isobars

B) Isotopes

C) Ions

D) Molecules

Answer: B

17. Atoms of different elements with same mass are

A) Isotopes

B) Isobars

C) Allotropes

D) Compounds

Answer: B

18. Different forms of same element are

- A) Isotopes
- B) Isobars
- C) Allotropes
- D) Ions

Answer: C

19. Carbon exists as

- A) Coal only
- B) Graphite only
- C) Diamond only
- D) Coal, graphite and diamond

Answer: D

20. Subatomic particles are

- A) Cells
- B) Protons, neutrons, electrons
- C) Atoms
- D) Molecules

Answer: B

21. Cathode rays were discovered by

- A) Rutherford
- B) Goldstein
- C) William Crookes
- D) Chadwick

Answer: C

22. Cathode rays originate from

- A) Anode

B) Cathode

C) Nucleus

D) Gas

Answer: B

23. Cathode rays are

A) Neutral

B) Positively charged

C) Negatively charged

D) Uncharged

Answer: C

24. Cathode rays travel in

A) Zigzag path

B) Circular path

C) Straight line

D) Random path

Answer: C

25. Cathode rays are deflected by

A) Heat

B) Light

C) Electric and magnetic fields

D) Gravity

Answer: C

26. Electron was discovered using

A) Gold foil experiment

B) Discharge tube

C) Electrolysis

D) Flame test

Answer: B

27. Anode rays are also called

A) Cathode rays

B) Canal rays

C) X-rays

D) Gamma rays

Answer: B

28. Anode rays were discovered by

A) Crookes

B) Thomson

C) Goldstein

D) Rutherford

Answer: C

29. Anode rays carry

A) Negative charge

B) No charge

C) Positive charge

D) Variable charge

Answer: C

30. Properties of anode rays depend on

A) Cathode material

B) Gas used

C) Battery

D) Glass tube

Answer: B

31. Neutron was discovered by

- A) Thomson
- B) Rutherford
- C) Chadwick
- D) Goldstein

Answer: C

32. Neutron has

- A) Positive charge
- B) Negative charge
- C) No charge
- D) Variable charge

Answer: C

33. Mass of neutron is approximately equal to

- A) Electron
- B) Proton
- C) Alpha particle
- D) Photon

Answer: B

34. Thomson proposed atomic model in

- A) 1808
- B) 1878
- C) 1904
- D) 1932

Answer: C

35. Thomson model is also called

- A) Nuclear model

- B) Planetary model
- C) Plum pudding model
- D) Wave model

Answer: C

36. Thomson model explained

- A) Nuclear structure
- B) Electrical neutrality
- C) Neutron presence
- D) Orbits

Answer: B

37. Rutherford model is known as

- A) Plum pudding model
- B) Nuclear model
- C) Solid sphere model
- D) Cloud model

Answer: B

38. Rutherford experiment used

- A) Alpha particles
- B) Beta particles
- C) Cathode rays
- D) Neutrons

Answer: A

39. Most of atom is

- A) Solid
- B) Liquid
- C) Empty space

D) Gas

Answer: C

40. Electrons revolve around nucleus in

A) Waves

B) Orbits

C) Bonds

D) Clouds

Answer: B

41. Outermost electrons are called

A) Core electrons

B) Free electrons

C) Valence electrons

D) Bond electrons

Answer: C

42. Stable configuration has

A) 1 electron

B) 4 electrons

C) 2 or 8 electrons

D) 10 electrons

Answer: C

43. Valency is the ability to

A) Gain mass

B) Lose or gain electrons

C) Conduct electricity

D) Reflect light

Answer: B

44. Loss of electrons gives

- A) Negative valency
- B) Zero valency
- C) Positive valency
- D) Variable valency

Answer: C

45. Gain of electrons gives

- A) Positive valency
- B) Negative valency
- C) Zero valency
- D) Double valency

Answer: B

46. Valency of hydrogen is

- A) 0
- B) 1
- C) 2
- D) 8

Answer: B

47. Valency of oxygen is

- A) 1
- B) 2
- C) 3
- D) 4

Answer: B

48. Valency of chlorine is

- A) 1

B) 2

C) 3

D) 4

Answer: A

49. Iron shows

A) Fixed valency

B) No valency

C) Variable valency

D) Zero valency

Answer: C

50. Charged atom is called

A) Molecule

B) Radical

C) Ion

D) Element

Answer: C

51. Positively charged ion is

A) Anion

B) Cation

C) Molecule

D) Isotope

Answer: B

52. Negatively charged ion is

A) Cation

B) Proton

C) Anion

D) Neutron

Answer: C

53. Na^+ is an example of

A) Anion

B) Neutral atom

C) Cation

D) Molecule

Answer: C

54. Cl^- is an example of

A) Cation

B) Anion

C) Radical

D) Molecule

Answer: B

55. Ion with charge 1 is

A) Divalent

B) Trivalent

C) Monovalent

D) Tetravalent

Answer: C

56. Ion with charge 2 is

A) Monovalent

B) Divalent

C) Trivalent

D) Tetravalent

Answer: B

57.Oxygen is

- A) Monovalent
- B) Divalent
- C) Trivalent
- D) Tetravalent

Answer: B

58.Nitrogen is

- A) Monovalent
- B) Divalent
- C) Trivalent
- D) Tetravalent

Answer: C

59.Chemical formula shows

- A) Shape of molecule
- B) Number of atoms
- C) Colour of compound
- D) State of matter

Answer: B

60.Positive radical is written on

- A) Right side
- B) Left side
- C) Below
- D) Above

Answer: B

61.Valencies are written as

- A) Subscripts first

B) Superscripts

C) Prefix

D) Symbols

Answer: B

62. Formula of magnesium chloride is

A) MgCl

B) MgCl_2

C) Mg_2Cl

D) Mg_2Cl_2

Answer: B

63. Suffix used for molecular compounds is

A) -ate

B) -ite

C) -ide

D) -ol

Answer: C

64. Prefix di means

A) One

B) Two

C) Three

D) Four

Answer: B

65. Symbolic representation of reaction is

A) Chemical formula

B) Chemical equation

C) Skeleton

D) Valency

Answer: B

66. Reactants are written on

A) RHS

B) LHS

C) Top

D) Bottom

Answer: B

67. Arrow in equation means

A) Equals

B) Gives

C) Plus

D) Minus

Answer: B

68. Upward arrow shows

A) Precipitate

B) Liquid

C) Gas

D) Solid

Answer: C

69. Downward arrow shows

A) Gas

B) Precipitate

C) Liquid

D) Heat

Answer: B

70. Unbalanced equation is called

- A) Balanced equation
- B) Molecular equation
- C) Skeleton equation
- D) Ionic equation

Answer: C

71. Balancing uses

- A) Changing formula
- B) Adding coefficients
- C) Removing atoms
- D) Changing symbols

Answer: B

72. Small numbers in formula are

- A) Changed
- B) Removed
- C) Never changed
- D) Doubled

Answer: C

73. Balanced equation gives

- A) Speed of reaction
- B) Heat change
- C) Quantitative information
- D) Conditions

Answer: C

74. Physical state is shown by

- A) Balanced equation

B) Chemical formula

C) Not shown

D) Arrow

Answer: C

75. Law of conservation of mass was given by

A) Dalton

B) Proust

C) Lavoisier

D) Rutherford

Answer: C

76. Law of conservation of mass states mass is

A) Created

B) Destroyed

C) Constant

D) Variable

Answer: C

77. Another name of conservation law is

A) Law of motion

B) Law of indestructibility of mass

C) Law of gravity

D) Law of volume

Answer: B

78. Law of constant proportions was proposed by

A) Dalton

B) Lavoisier

C) Proust

D) Gay Lussac

Answer: C

79. Law of constant proportions deals with

A) Volume

B) Mass ratio

C) Temperature

D) Pressure

Answer: B

80. Water always contains hydrogen and oxygen in ratio

A) 2:1

B) 1:1

C) 1:8

D) 8:1

Answer: C

81. Laws of chemical combinations are based on

A) Colour

B) Shape

C) Quantitative measurements

D) Smell

Answer: C

82. Matter can neither be created nor destroyed during

A) Chemical change

B) Physical change

C) Both changes

D) Only nuclear change

Answer: C

83. Isotopes differ in

- A) Atomic number
- B) Chemical properties
- C) Mass number
- D) Valency

Answer: C

84. Isobars have same

- A) Atomic number
- B) Mass number
- C) Chemical properties
- D) Electrons

Answer: B

85. Allotropes differ in

- A) Atomic number
- B) Mass
- C) Properties
- D) Composition

Answer: C

86. Standard model deals with

- A) Planets
- B) Subatomic particles
- C) Cells
- D) Molecules

Answer: B

87. Discharge tube works at

- A) High pressure

B) Atmospheric pressure

C) Very low pressure

D) Vacuum only

Answer: C

88. Electron carries

A) No charge

B) Positive charge

C) Negative charge

D) Double charge

Answer: C

89. Proton is present in

A) Orbit

B) Nucleus

C) Shell

D) Outside atom

Answer: B

90. Neutron is present in

A) Orbit

B) Nucleus

C) Shell

D) Outside atom

Answer: B

91. Valency of magnesium is

A) 1

B) 2

C) 3

D) 4

Answer: B

92. MgO shows valency of magnesium as

A) 1

B) 2

C) 3

D) 4

Answer: B

93. Sodium chloride shows valency of sodium as

A) 1

B) 2

C) 3

D) 4

Answer: A

94. Chemical compounds have properties

A) Same as elements

B) Different from elements

C) Same as metals

D) Same as non-metals

Answer: B

95. Prefix mono means

A) One

B) Two

C) Three

D) Four

Answer: A

96. Chemical equation does not show

- A) Reactants
- B) Products
- C) Speed of reaction
- D) Ratio of atoms

Answer: C

97. Gay Lussac law deals with

- A) Mass
- B) Volume of gases
- C) Temperature
- D) Pressure

Answer: B

98. Law of constant proportion depends on

- A) Source
- B) Method
- C) Fixed mass ratio
- D) Surface

Answer: C

99. Electron arrangement is called

- A) Valency
- B) Atomic number
- C) Electronic configuration
- D) Mass number

Answer: C

100. Stability of atom is achieved by

- A) Maximum mass

B) Complete outer shell

C) Maximum charge

D) More neutrons

Answer: B